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Structural analysis and antioxidant activities of polysaccharides from cultured *Cordyceps militaris*

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Abstract

The polysaccharides extracted by hot water and precipitated by 50% ethanol from the fruiting bodies of cultured *Cordyceps militaris* (W-CBP50) were introduced. Antioxidant activities of W-CBP50, W-CBP50 I (obtained from W-CBP50 purified by Sephadex G-100 chromatography) and W-CBP50 II (obtained from W-CBP50I purified by Sephadex G-200 chromatography) were evaluated and structural features of W-CBP50 II were analyzed. The polysaccharide W-CBP50 II was able to scavenge the stable free radical 1,1-diphenyl-2-picrylhydrazyl (DPPH), while W-CBP50 and W-CBP50I could scavenge DPPH, hydroxyl and superoxide radical. Structural characteristics of W-CBP50 II were investigated using GPC, GC-MS, (1)H NMR and IR. The results showed that it was mainly composed of α -glucose, α -mannose, α -galactose and α -arabinose with α -type glycosidic linkage and the molecular weight of its main component was 8.97 kD.

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